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BRING YOUR ONW DEVICE

Ch. Cuvelliez – Feb 11

BYOD

AGENDA

- **Definition**
- **Objectives**
- **Risk of BYOD**
- **Alternatives to BYOD: Why Not**
- **Develop the policy**
- **Enforce Controls**
- **Deployment Approaches**
  1. Web browsers
  2. Virtual Desktop Infrastructure (VDI)/Remote Desktop/Remote Apps)
  3. Bootable OS
  4. Mobile Device Management (MDM
  5. Mobile Application Management (MAM)

BYOD

DEFINITION

BYOD is the concept of employees using their personally owned device(s) for work purposes.

With BYOD, an organization has **ownership of the corporate data and resources that may be accessed or stored on a device**: device itself is the property of the user.

- Give end-users the **ability to use IT they feel comfortable** with
- Reduce **overheads for the organization in terms of procurement and provisioning of** corporate devices
- Enable flexible (including remote) working
- **Increase productivity**
- Provide **redundancy to business and organizations when workers** are unable to access their main places of work

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BYOD

OBJECTIVES

Is BYOD intended to be an interim or long-term solution?

With BYOD, an organization has **ownership of the corporate data and resources that may be accessed or stored on a device**: device itself is the property of the user.

- **Short-term solutions often start with the right intentions but rapidly becomes long-term** implementations: then it is not fit for purpose, but too late, difficult to remove, especially if used by a large volume of staff.

- If you decide this is purely a short-term solution, **define an end date before implementation and do not deviate from it.**

BYOD

OBJECTIVES

What business functions need to be achieved with BYOD?

- For example, minimal ‘keeping in touch’ facilities like calendars will present a lower risk than additional functionality like access to office functions such as documents and file sharing.

- For example performing remote system management should not be possible with BYOD

What types of devices do you intend to facilitating work from?

- **Smartphones, laptops, desktop PCs, tablets**, or a combination ?
- Better **to start with one category and then expand** the deployment over time.
- Also **define at a more granular level what platforms** (and versions) you will/can support: iOS, Windows, Android, macOS.
- Useful to survey what devices and platforms are currently used by your BYOD workforce.

BYOD

OBJECTIVES

What business functions need to be achieved with BYOD?

- For example, minimal ‘keeping in touch’ facilities like calendars will present a lower risk than additional functionality like access to office functions such as documents and file sharing.

- For example performing remote system management should not be possible with BYOD

Will this be your only flexible working solution and where will BYOD devices be used?

- Mix of BYOD and Corporately Owned devices used for flexible working?
  - Some users may not be able to afford devices capable of BYOD, so a hybrid model of both BYOD and Corporately Owned/Managed to be considered if flexible working is required.
- Will **BYOD only be used for remote working**, or also brought **into office spaces for flexible working and connected to ‘trusted’ networks?**

BYOD

OBJECTIVES

User Needs

- Identify unsuitable BYOD users based on data sensitivity and risk.
- Executives handling confidential data (e.g., financials, PII, IP) should be excluded or restricted to minimal access via secure enclaves/containerization.

Unsuitable Groups

- **Executives/C-Suite**: High-sensitivity data exposure risk; mandate corporate devices.
- **HR/Finance/Legal**: PII, payroll, contracts—too critical for personal device mixing.
- **R&D/Engineering**: Proprietary code/IP; full exclusion preferred.
- **High-privilege IT admins**: Elevated access amplifies breach impact.

Suitable Groups (Pilot)

Involve these early in development for feedback:

- **General office staff**: Email, docs, low-sensitivity apps.
- **Sales/Marketing**: CRM access, client comms (non-confidential).
- **Remote/field workers**: Basic productivity tools.

Conflicting priorities

The priorities and desires of users and organisations differ: involve users in the BYOD Strategy !

- Usability = focus of owners (no disruption of their usual experience of a device+ concerns over the privacy of their personal data + trust of corporate control you intend to implement.
- Security and integrity = focus of the organization

If one party is disappointed, BYOD will be abandoned

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RISK OF BYOD

Risk (1/3)
Easier user-initiated deliberate loss of data, eg copying data from work to personal apps
Higher potential accidental data loss, eg device backups containing work data, users sharing their device with family
Malicious exfiltration of data, eg malicious applications leaking data that users have allowed it to access
Less trust in a BYOD device at the point of enrolment/first use : • **Pre-existing Device History**: BYOD devices likely had personal use before enrollment, making their initial "health state" unknown. • **Unreliable Reporting**: No assurance the device accurately reports details like OS version or security status during first business use. • **Enrollment Risks**: Potential for hidden issues (malware, outdated software) that self-reporting cannot verify reliably.

Risk (2/3)

Workers having access to more resources and services than required

- **Excessive Access Risk:** Workers often get more resources/services than needed, violating least privilege.
- **Service Suitability:** Not all corporate services fit BYOD—limit to aligned objectives, needs, and low-risk subsets vs. full office access**.**
- **Tailored per Role:** Customize resources by user/group (e.g., finance vs. engineering apps) to minimize data exposure.
- **Offboarding Concerns:** Hard to ensure complete removal of corporate data/apps/credentials from BYOD devices upon employee exit/termination.

Higher likelihood of unsupported or out of date devices, leading to exploitation of known security vulnerabilities by commodity malware

Additional exposure of devices to threats because they are being used in a broader personal context, such as users sharing devices or passwords with others

Users being less willing to report security incidents involving their personal device to an organisation for fear of intrusion into personal data.

Risk (3/3)

Malicious exploitation of devices because of weak security configuration, for example no data at rest encryption, allowing data extraction or a lack of physical security controls on the device, such as Passwords/PIN/biometrics

Malicious exploitation of devices remaining undetected due to lack of monitoring. Potentially leading to further spread of malware such as screen scraping and keyloggers, resulting in exfiltration of credentials and corporate data.

Increased risk of device theft, loss and breakage

- **Increased Transport Risk**: BYOD devices travel more frequently with users (personal habit vs. corporate), raising loss/theft exposure.
- **Untrusted Repairs**: If broken, users may use non-reputable repair shops, risking corporate data access/exfiltration.
- **Cost/Insurance Dilemma**: Decide who pays for repairs/replacements during work use—user or organization—and handle insurance implications.

Use of unsecure networks and public locations, leaves risk of eavesdropping to access corporate resources (eg, public transport, cafés etc.)

Infected devices, for example being taken into the office and connected to internal networks.

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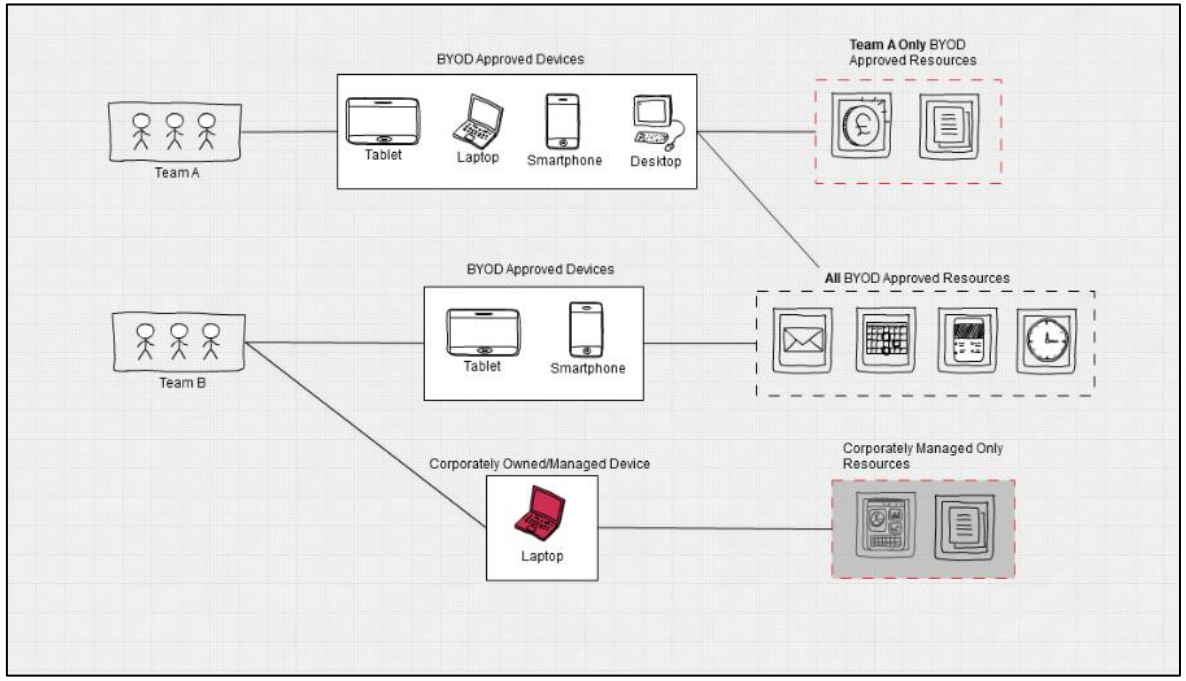
ALTERNATIVES TO BYOD: WHY NOT

- **Corporately Owned**, Business Only,
- **Corporately Owned**, Personally Enabled,
- **Choose Your Own Device**,
- **Personally Owned, Corporately Managed** – where full device management is given to the organisation.



No 'one size fits all' BYOD solution.

- **Broad BYOD Access**: All workers to use BYOD for viewing management comms, timesheets, team/HR instant messaging on smartphones, tablets, laptops, or PCs.
- **Limited Corporate Features**: Select users BYOD access to corporate email and non-sensitive documents for read/write/edit.
- **Sensitive Apps Exclusion**: Certain department users need remote-suitable apps too sensitive for BYOD, requiring corporately owned/fully managed devices instead.

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DEVELOP THE POLICY

- **What tasks will employees be permitted to do from their own devices?**
  Users with privileged or administrative access should not have that access from a personal device

- **What services expose to personal devices? What data expose from within those services?**
  e.g. Allow users to submit expense reports, but not permit changing of bank details.

- **How much control will your employees be willing to grant you over their devices, and how much control do you need?**
  If users rejects any control of their devices, managing risks will be compromised. E.g., Users may not like the idea of their employer being able to remotely wipe their entire device.

- **How enforceable are your policies?**
  What happens if users do not follow those procedures,
- **Will employees be able to use more than one BYOD device? Where will BYOD devices be used? How to manage a security incident?**

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ENFORCE CONTROLS

- **What types of client access are permitted?**
  For example, phones with native apps, third-party container apps, even web browsers or plug ins

- **What minimum standards for hardware and software versions will you enforce?**
  Older or unsupported platform versions are more likely to contain security weaknesses for which no fix exists

- **What policies will you enforce, and how will you enforce them? Will they apply at device, application or both levels?**
  You may be able to enforce some policies, including minimum passcode length and preventing copy and paste between work and personal apps

- **How will you handle an employee changing their device, changing their role, or leaving the organisation?**
  What method will you use to remove access and corporate data (if stored on the device)?

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ENFORCE CONTROLS

- **What service access policies will you enforce?**
  For example, you could use compliance policies and strong authentication to verify devices before they are allowed access to BYOD approved enterprise services.

- **How will individual services prevent personal devices from accessing sensitive data?**

- **How and where will you enforce these policies?**
  They could be enforced at an authentication service, network firewall, or boundary/perimeter of specific services.

- **How will you enforce separation between business and personal applications?**
  What will you use to prevent corporate data being opened or moved to personal applications? For example, you may use MAM policies to block copying of corporate data to personal storage services.

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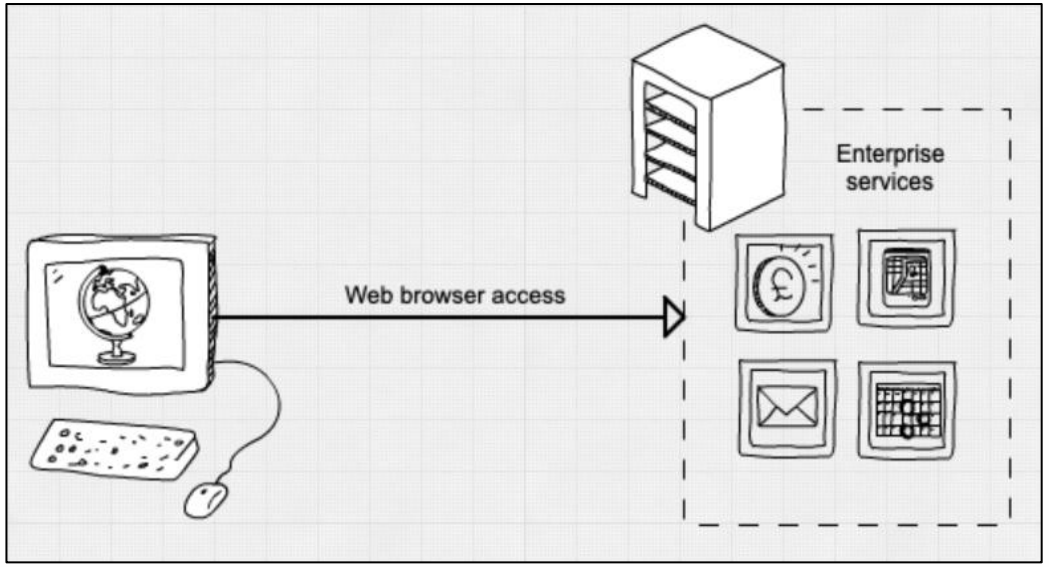
DEPLOYMENT APPROACHES

Web browsers

You may consider allowing personal devices to access corporate data through the web browser

- **Webmail BYOD Approach**: Commonly used for employee access to work email due to its platform-agnostic simplicity and versatility.
- **Risks of Simplicity**: Lacks enforceable technical controls against data loss or insecure service access; no assurance of device security/configuration.
- **Compromise Impact**: Attacker on compromised device can easily access rendered data, browser cache, credentials, and service tokens.
- **Mitigation Focus**: Educate employees on risks and underlying reasons ("why"), not just procedures, to ensure they value security guidance.



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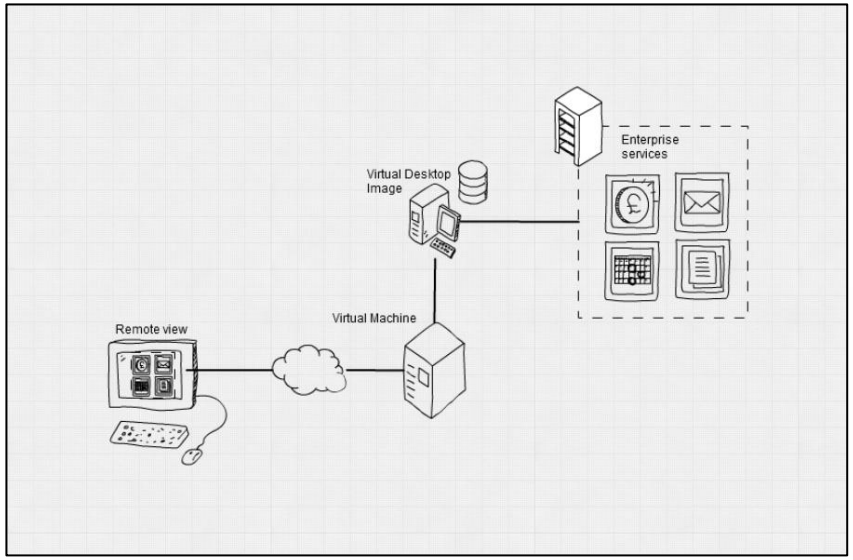
DEPLOYMENT APPROACHES

Virtual Desktop Infrastructure (VDI)/Remote Desktop/Remote Apps

Remote view = minimal amounts of corporate data are cached on the user device

- **Limited Data Access**: Controlled corporate data view without direct file exposure on BYOD devices.
- **Malware Risks Persist**: Screen scraping, keylogging (credentials), or keystroke/mouse injection can still capture data or compromise remote sessions.
- **Acceptable for Low-Sensitivity**: Fine if users access only non-sensitive info; MFA mitigates keylog credential theft.
- **Data Exfiltration Reduced**: No bulk file transfer risk, though injected actions could enable remote compromise.
- **Minimal Local Storage**: Little corporate data on endpoint; loss/theft impact low (remote VDI wipe possible).
- **Pre-Access Checks**: Can verify AV presence; strong auth enforced, but doesn't detect full compromise.
- **Focus on Corporate Controls**: Secure the managed VDI asset centrally—enforce policies blocking data copy/rendering outside virtual env.



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DEPLOYMENT APPROACHES

Virtual Desktop Infrastructure (VDI)/Remote Desktop/Remote Apps

Pros

- **Minimal amounts of corporate data stored** on the device
- **Less corporate data can be compromised** in well configured virtual environments
- **Include controls to lower the risk of exfiltration of corporate data** to personal applications and environments
- **Provide simple user device and app compliance checks**
- Can require strong authentication for access
- **Focus is on securing the corporate infrastructure and resources that you can control, rather than the user device that you cannot**
- Data collected about the user's device can help enforce access decisions

Cons

- Devices **still vulnerable to malware and controls are limited if the host is compromised**
- **Poorly configured RDP solutions are an attractive target** for ransomware attacks
- **Dependant on a strong and stable connection to remote environment compared to browser access**, which can cache and recover from temporary connection outages.

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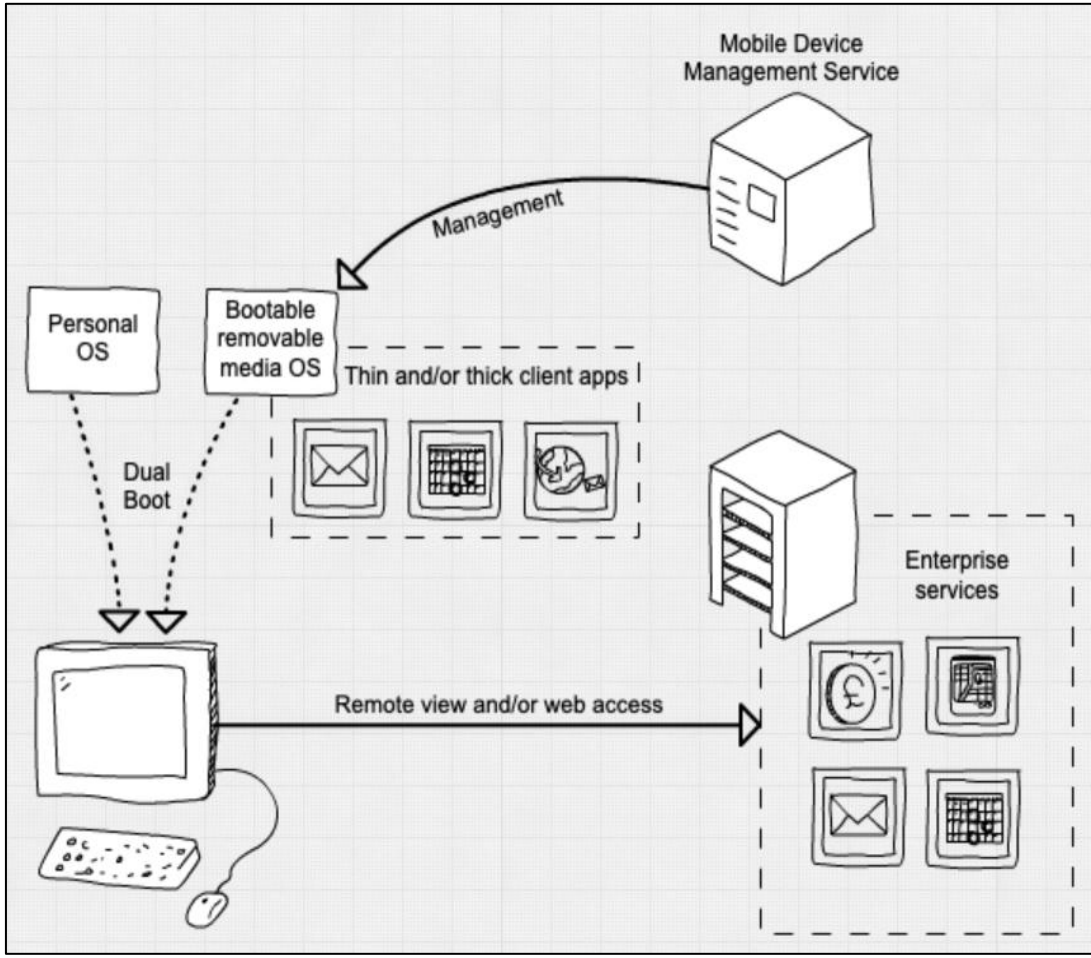
DEPLOYMENT APPROACHES

Bootable OS

- Boot the user's device into a managed operating system environment **using bootable media, typically USB**.
- **Variety of third-party products** to do this, as well as Windows to Go, which is built into some older versions of Windows
- The **lowest risk way of enabling home PCs** to be used but:
  - How to **get confidence in the underlying hardware**.
  - Firmware attacks on the host device are still possible, + other risks, such as not being able to follow best practice on storing keys



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DEPLOYMENT APPROACHES

Bootable OS

Pros

- Organisations can **manage the applications available to the device when provisioning the bootable media**
- Lowest risk way of enabling home PCs for BYOD.

Cons

- Less accessible due to **complexity of setup**
- Potentially **requires reconfiguration of the personal device's firmware**
- Managed operating system is still **reliant on the underlying device hardware**
- **Trust** in the device is **not solely based on the bootable operating system**.

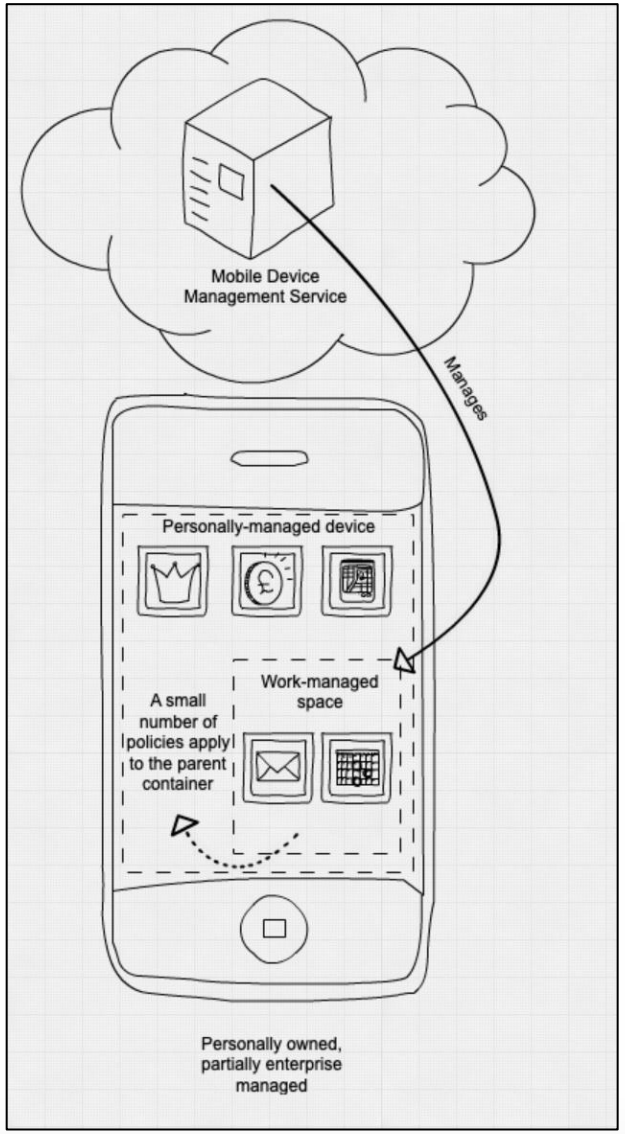
BYOD

DEPLOYMENT APPROACHES

Mobile Device Management (MDM

Using Mobile Device Management (MDM) for BYOD involves enrolling personally owned devices in a corporate solution.

- This enrollment **grants the enterprise a degree of control over the device and its settings.**
- The model is "partly managed."
- All modern device platforms support MDM, including iOS, macOS, Android, ChromeOS, and Windows 10.
- Each platform offers differing levels of security controls.
- MDMs are usually able to enforce some device-wide configuration policies, as well as policies which protect corporate data within apps, or managed accounts



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DEPLOYMENT APPROACHES

Mobile Device Management (MDM

Monitoring Capabilities:

Report a wide range of device states, for establishing the health of the device.

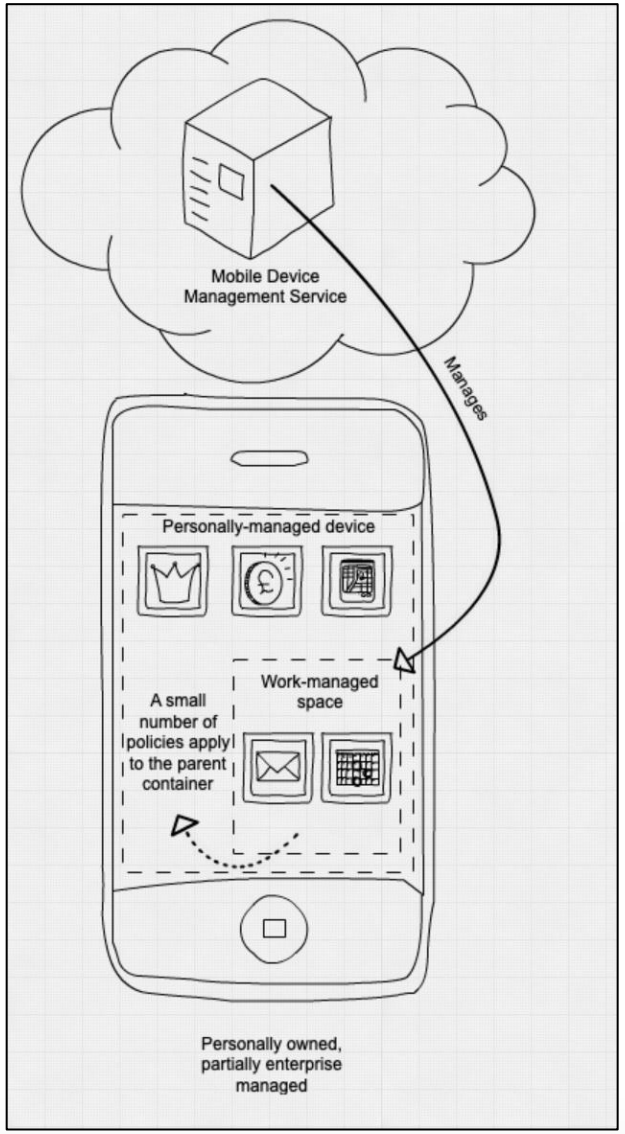
- The information **presented to the organisation originates from the MDM and not the actual events on the device** (blocked by MDM protocols to respect privacy)
- A **compromised personal device may also obscure or conceal its activity** from MDM reporting.

Data which MDMs can report include:

- A list of apps installed on the device
- Are anti-malware features enabled
- Can apps be installed from less trusted sources
- Device unlock settings
- Is device encryption enabled

Some may even include information such as:

- MAC address
- Phone number
- Current and previous device names



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DEPLOYMENT APPROACHES

Mobile Device Management (MDM

Pros

- **Greater assurance in the device than MAM-only** (see below)
- **Security controls implemented and enforced at the platform level.**
- Strong authentication can be enforced along with **access controls based on perceived device**.
- Access control policies can be configured to **require that a device is registered + configuration, patch status and malware mitigation availability** is compliant.
- Can prompt the users to ensure that their device and browser have received recent updates.
- Will **push users to download work apps from a corporate store app**.

Cons

- Provides levels of device controls and device information not all appropriate for BYOD.
- Corporate data and profiles will be stored on the device + Stronger controls could be rejected by the users.
- Data used to enforce security controls is only as good as the integrity of the underlying platform.

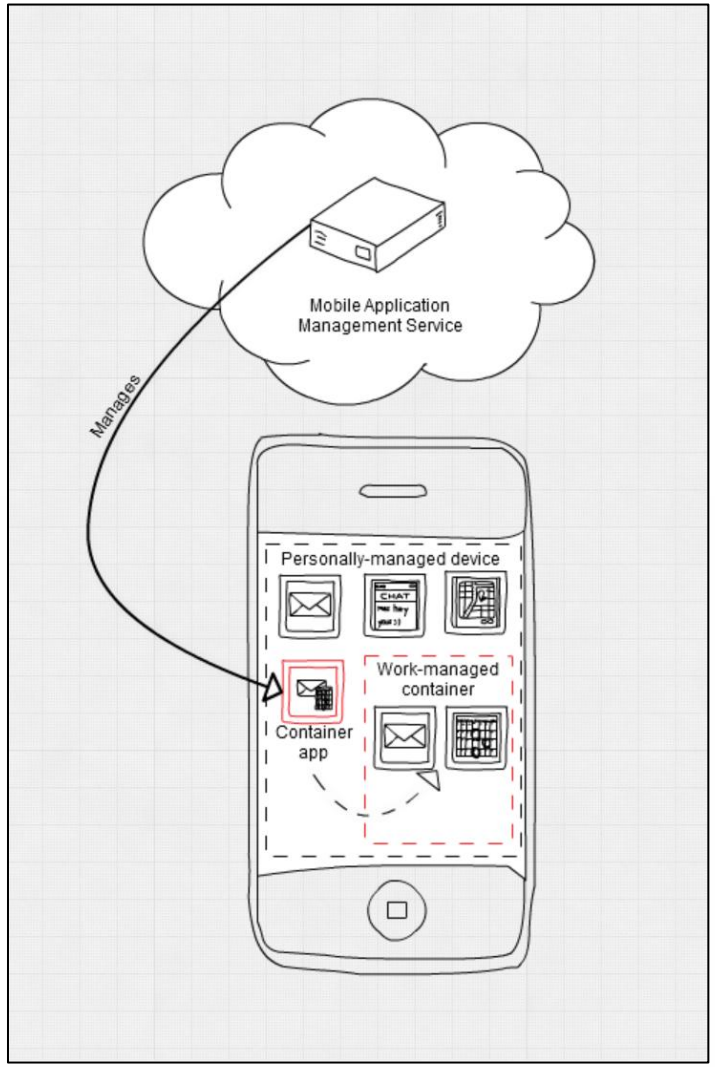
BYOD

DEPLOYMENT APPROACHES

Mobile Application Management (MAM)

The user manages all aspects of the device, except for work applications, which are held in a container on the device and managed by the organization

- The **organization owns all data and resources stored within the containers**.
- Corporate administrators **can remotely apply settings to MAM (Mobile Application Management)-enabled apps installed in the container**.
- **Security controls are enforced at the app level**, not by the device's operating system or platform.
- **Poorly configured or compromised devices can weaken these app-level security** measures.
- The container model **limits control over the full device** but **enhances protection and separation of corporate data from personal apps.**
- **Container apps often block risky actions like copy-paste** or screenshots **between corporate and personal areas.**
- Some **MAMs can monitor device details, such as OS version, though this data is less reliable without full device management**.



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DEPLOYMENT APPROACHES

Mobile Application Management (MAM)

Pros

- **Corporate controls are restricted to organisation owned data and managed services** only
- **Restrictions on the transfer of data out of the containers can be implemented**
- Authentication and access controls checks can be implemented
- **PIN requirements can be enforced to unlock managed app(s)**
- **Remote wiping or access revocation is limited to corporate data and managed container applications only**
- **Likely to be more appealing to the end user due to the separation of work and personal security controls**.

Cons

- **Per-app security controls are heavily reliant on good configuratio**n
- **Malware infected devices can gain access to corporate data, personal data, and security tokens**
- **Compromised devices can spoof information to the managed app**, undermining its security controls
- **User's device may have reduced functionality.**
  - For example, built-in mail and calendar applications may not be able to sync with a corporate app
- Many of the risk mitigations with MAM-WE are only effective on well configured, patched devices.